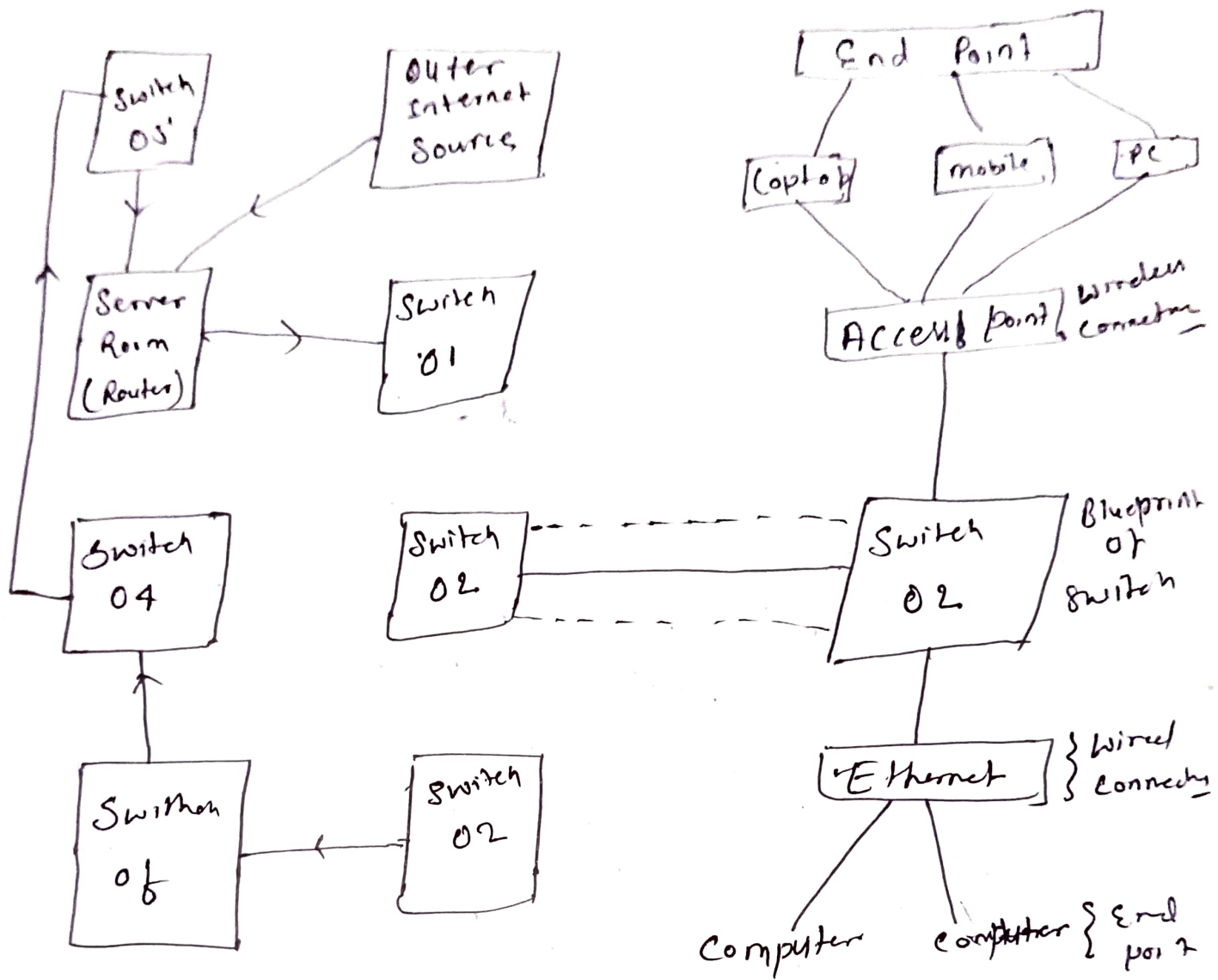


1. Draw the Network topology:-

(3)



- Combination of Ring & Star topology.

2. Network devices used in LAN.

(1) Router:-

Function:- connects completely different network to gather such as connecting your private home / college LAN to the public internet.

5. Find system network n.t. in.

(ii) Network Switch :-

Function :- connects multiple end-devices (Pcs, printers, servers) together within the same LAN.

(iii) Wireless Access Point (WAP) :-
Function :- allows wifi-enabled devices to connect to a wired network.

(3) Analyze connection types.

* Wireless :- Devices such as mobile phones, laptop, PC, which are connected via access point.

* Wired :- Devices such as computer, smart boards, PC which are connected via access point. Ethernet.

(4) Measure network speed:

- Commands or tools which are used to measure the linked speed of wired network and wireless networks are:-
Wired network :- network speed (from terminal)
Wireless network :- network speed (from terminal)

5. Find system network details:

(37)

- Commands used are:-
for macbook:-

MAC address

IP address

Subnet mask

} if config en's.

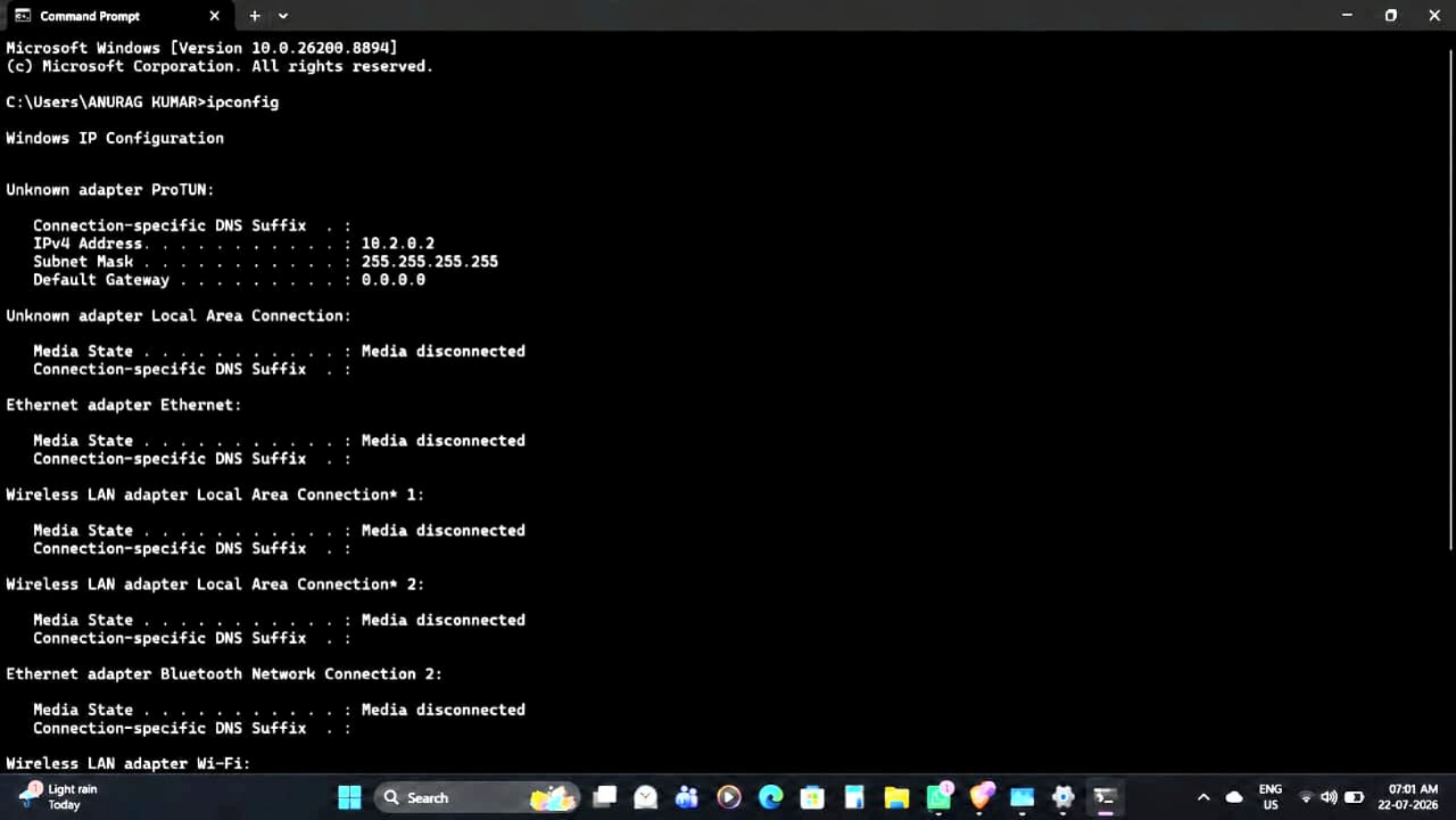
6. Explain observations:

① Difference between wired and wireless speed:-

- Wired speed is faster, more stable and has lower latency than wireless because data travels through isolated physical cables, rather than open air, while modern wireless technology boasts high theoretical speed, real world faster came it to full short of a direct wired connection.

② Role of subnet mask in networking:-

- A subnet mask separates an IP address into two parts: The network address and the host address, it all act like a filter that helps router and devices determine if a destination IP address is on the same local network or outside of it.



```
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ANURAG KUMAR>ipconfig

Windows IP Configuration

Unknown adapter ProTUN:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 10.2.0.2
    Subnet Mask . . . . . : 255.255.255.255
    Default Gateway . . . . . : 0.0.0.0

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Bluetooth Network Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:
```

